

ptt-tmux

Push-to-talk (toggle-to-talk) voice input for tmux. Speak into a terminal pane and have your words typed in live as you say them, powered by Parakeet running locally via MLX.

How it works

- **daemon.py** loads the Parakeet model once and listens on a unix socket (`~/.ptt-tmux.sock`). While recording is active it re-transcribes the audio-so-far roughly once a second and types only the newly recognized words into the target tmux pane via `tmux send-keys`.
- **toggle.py** is a tiny client: it sends `toggle <pane-id>` to the daemon. First call starts recording, second call stops it.

Setup

Dependencies are already installed in `.venv` (`parakeet-mlx`, `sounddevice`, `numpy`). If you need to reinstall:

```
cd ptt-tmux
python3 -m venv .venv
.venv/bin/pip install parakeet-mlx sounddevice numpy
```

Try it out

1. Start the daemon (first run downloads the model, then loads it — takes a bit):

```
cd ptt-tmux
.venv/bin/python daemon.py
```

Leave this running in its own terminal/pane and watch its log output (`[ptt] ...`) for status.

2. In another pane, find your pane id:

```
tmux display-message -p '#{pane_id}'
```

3. Toggle recording on, targeting that pane:

```
.venv/bin/python toggle.py %<pane-id>
```

Start speaking. Words should appear typed into the target pane within about a second of saying them.

4. Run the same command again to stop recording:

```
.venv/bin/python toggle.py %<pane-id>
```

Wiring up a tmux key binding

Add to `~/.tmux.conf` so a single key toggles recording for whichever pane is focused (example binds it to `prefix + v`):

```
bind-key v run-shell "~/.GenAI/genai-workspace/ptt-tmux/.venv/bin/python
↪ ~/GenAI/genai-workspace/ptt-tmux/toggle.py #{pane_id}"
```

Reload with `tmux source-file ~/.tmux.conf`, then press `prefix + v` to start/stop recording in the active pane.

Notes / current limitations

- This is a first working pass, not optimized: transcription re-runs on the full audio buffer every poll interval (~1s), so latency and CPU usage grow with utterance length. Fine for short dictation; a longer session would want proper streaming/chunked decoding.
- Only one recording session at a time (single global daemon state).
- The daemon must already be running before you call **toggle.py** (**ping** command exists for a basic liveness check if needed).